

The Impact of Large-Scale Data Center Load Integration into Electrical Power Networks

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Abstract—With the rapid advancement of information and communication technology, data centers have experienced exponential growth, resulting in a substantial increase in their electrical load demand. It is projected that by 2030, global data center electricity demand could reach 321 TWh, assuming current growth trends persist. As large-scale data centers continue to expand, their integration into the power grid poses significant challenges to grid stability and reliability. This research investigates the impact of large-scale data center load penetration on electrical power networks, with a particular focus on frequency and voltage dynamics and operational integrity. Using historical data and detailed PSCAD simulations based on the IEEE 14-bus test system, this article presents a comprehensive analysis of how increasing data center loads affect grid stability. The results reveal that sudden high-capacity load fluctuations and disturbances can lead to frequency deviations, voltage instability, and increased stress on the electrical power network. Various operational scenarios are examined, and strategic recommendations are proposed to mitigate adverse effects and enhance the resilience of future power grids facing high data center penetration.

Keywords—Data center load, frequency stability, inverter-based resources (IBRs), load regulation, power grid stability, voltage stability.

I. INTRODUCTION

Over the past few decades, advancements in information and communication technology (ICT) have driven a rapid increase in computational tasks, including cloud computing, data processing, and storage. Consequently, the demand for large-scale data center infrastructure, encompassing cooling systems, data storage units, and uninterrupted power supplies (UPS), has risen substantially. As reported in [1], the total energy consumption of data centers increased modestly by 6% between 2010 and 2018, from 194 TWh to 205 TWh. However, projections indicate a significant increase in future energy demands, with estimates suggesting that global data center electricity consumption could reach 321 TWh by 2030 if current growth trends continue [1], [2]. The global electricity demand for data center load can be illustrated in Fig. 1.

At the same time, due to the decrease of fossil fuel-based energy sources, the influence of renewable energy sources such as solar and wind energy is becoming more popular day by day. Therefore, the number of wind and photovoltaic installations has been rising constantly. In the past 20 years, from 1997 to 2016, the global installed capacity of wind power increased from 7.64 GW to 468.99 GW, and the installed capacity of photovoltaic power increased from 0.23 GW to 301.47 GW, increasing by 60

times and 1284 times, respectively [1], [2]. The data center and renewable energy sources are commonly identified as inverter-based resources (IBRs), which require multiple inverters/converters.

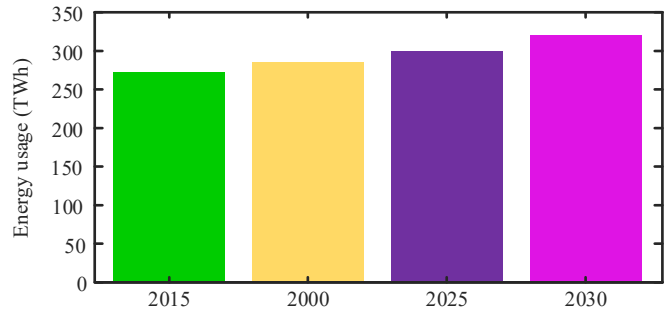


Fig. 1. Data center energy demand.

As IBRs become increasingly integrated into modern electrical networks, their large and often variable profiles pose critical challenges to power system stability and reliability. The integration of large-scale data center loads can influence key grid parameters such as bus voltages, branch power flows, and system frequency, thereby impacting overall grid performance [3]. In extreme cases, power flow through grid branches may exceed design limitations, potentially resulting in overheating or fire hazards, which pose serious risks to safe grid operation [4]. Additionally, unregulated frequency deviations due to abrupt load changes can lead to system instability, frequency collapse, and even large-scale blackouts if not managed promptly [5].

These emerging challenges highlight the pressing need for intelligent load regulation strategies to ensure the secure and reliable integration of data centers into the power grid. A core issue lies in maintaining critical grid parameters, voltage, frequency, and power factor within acceptable thresholds, collectively referred to as the power grid parameters.

Motivated by these concerns, this article investigates the operational impact of large-scale data center loads on electrical power networks and proposes a dynamic regulation strategy tailored to mitigate their adverse effects. The key contributions of this work are summarized as follows:

- Modeling of large-scale data center loads, with analysis of their sudden load demand change on the normal operation and overall stability of the grid;
- Evaluation of the impact of large-scale data center loads on grid stability during network faults; and

- Formulation of practical recommendations for characterizing and managing data center operations to enhance grid stability under realistic operating conditions.

The remainder of this article is structured as follows: Section II reviews the relevant literature, and Section III presents the simulation setup in the PSCAD environment and highlights the key factors. Section IV represents the regulation strategies for data center load connection. Finally, Section V summarizes the main findings and outlines potential directions for future research.

II. LITERATURE REVIEW

This section presents the literature review and background on data center load, focusing on its key characteristics, the challenges and impacts of integrating such loads (under sudden load changes and fault conditions) into power grids, and the existing policies governing their grid connection.

A. Characteristics of data center load.

Data centers are specialized facilities designed to house computing infrastructure for digital services such as cloud computing, storage, artificial intelligence (AI), and real-time data processing. One of the defining characteristics of data center loads is their high-power density, with power demands ranging from a few megawatts in small-scale centers to over 100 MW in hyperscale facilities [6], [7]. Unlike conventional loads, data centers exhibit continuous, mission-critical, and non-interruptible demand profiles, with operations often maintained 24/7 to ensure service reliability.

Another notable feature is the low load variability at short timescales, which can shift dramatically during peak processing periods such as AI model training or cloud traffic surges. These loads are also characterized by a low power factor and a high prevalence of non-linear power electronics, such as UPS, rectifiers, inverters and variable-speed drives in cooling systems, which can introduce harmonics and other power quality disturbances into the grid [8]. Existing literature has also noted that cooling systems alone can account for 30–40% of a data center's total power consumption, further influencing load patterns [7], [8].

B. Challenges and impact of data center load integration into power grids

The large-scale integration of data center loads into electrical power networks presents multiple technical challenges, particularly in the domains of voltage regulation, frequency control, and power quality.

Voltage and frequency instability: Due to their large and concentrated power demands, data centers can cause significant voltage drops in local distribution networks, particularly in weak or rural grids [2]. In areas where renewable energy penetration is high, their relatively inflexible demand can exacerbate frequency fluctuations and reduce overall grid stability [9]. Studies suggest that the ramping behavior of data centers, especially during backup transitions or equipment failure, can cause transient disturbances affecting frequency stability [9].

Power quality issues: The widespread use of power inverter-based resources (non-linear characteristics due to the highly inductive load) in data centers leads to the generation of harmonics, voltage flicker, and imbalanced loads, which degrade power quality both within the facility and in the surrounding grid [10]. Harmonics, in particular, can lead to increased losses in transformers and conductors, overheating of equipment, and malfunction of sensitive devices.

Reactive power and grid congestion: Data centers often exhibit a low net power factor due to the inductive and capacitive nature of internal equipment, increasing reactive power demand from the grid. Furthermore, clustering of large data centers in specific geographic regions has led to localized grid congestion, necessitating major infrastructure upgrades and new substation installations [10], [11].

In response, researchers have proposed a range of solutions, including the deployment of on-site energy storage, grid-supportive inverter controls, and demand-side management strategies to mitigate these impacts [11].

C. Existing policies related to the data center load connection to the grid

Given the critical importance of maintaining grid reliability, several standards and policies have been established to regulate how data centers are connected to the grid and how they manage their power consumption.

Voltage, frequency, and power quality compliance: Regulatory authorities such as the Institute of Electrical and Electronics Engineers (IEEE) and International Electrotechnical Commission (IEC) have developed technical standards like IEEE 519 (harmonic limits), IEEE 1547 (interconnection standards), and IEC 61000 (electromagnetic compatibility), which are mandatory or recommended for large-scale loads like data centers (IEEE, 2018; IEC, 2021) [12], [13]. These standards enforce strict compliance to ensure that grid-connected equipment does not compromise voltage and frequency stability [12], [13].

Grid code requirements: National grid operators, such as the Australian Energy Market Operator (AEMO) and the North American Electric Reliability Corporation (NERC), have issued grid codes requiring large customers to support grid stability through voltage ride-through, frequency response, and power factor control (AEMO, 2023; NERC, 2021) [12], [13].

Energy efficiency and sustainability initiatives: In addition to stability-focused policies, many governments have introduced energy performance targets and sustainability certifications for data centers. These include mandates for power usage effectiveness (PUE) reporting, integration of renewable energy sources, and participation in demand response programs to reduce peak demand impacts [12], [13].

Despite these policies, literature highlights a gap in region-specific planning frameworks, particularly in fast-growing data center markets where grid capacity may lag behind demand growth. This underscores the need for proactive planning and real-time monitoring strategies to ensure reliable integration.

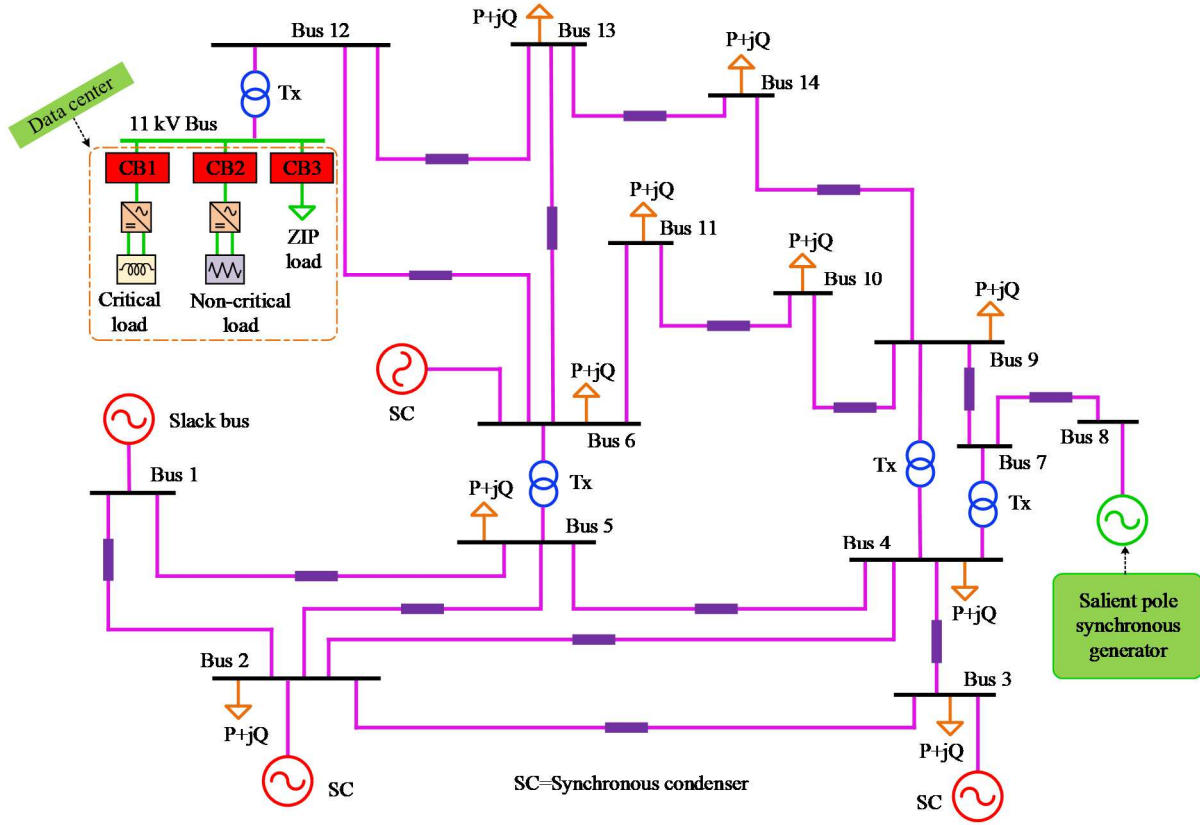


Fig. 2. IEEE 14-bus test system with modification.

III. SIMULATION SETUP AND RESULTS ANALYSIS

In this section, the simulation setup developed in PSCAD v50 with time step of 5 μ s, a widely used industry-oriented software for power system analysis, is presented along with the corresponding results. The work emphasizes the critical aspects of data center behavior that directly influence power system stability under different operating conditions. Key performance indicators, including frequency variation, voltage response, and power flow characteristics, are examined to assess the system's dynamic performance. The obtained results provide valuable insights into the stability margins of the test system, thereby validating the effectiveness of the proposed modeling approach and highlighting its practical relevance for real-world power system applications.

For the stability assessment, the IEEE 14-bus test system has been employed as the benchmark network as presented in Fig. 2. The modified terminal conditions are presented in Table I, and transmission line parameters of test bus system used in this work are detailed in [14]. In this modified configuration, Bus 1 is modeled as an infinite bus, and the synchronous condenser originally connected at Bus 8 in [14] has been replaced with a 300 MVA salient pole synchronous generator representing a hydroelectric plant. The generator model is equipped with an exciter, power system stabilizer, multimass system, hydro governor, and hydro turbine to capture the dynamic behavior of hydroelectric generation. Further technical specifications of the synchronous generator model are provided in [15].

The data center load is integrated at the 11 kV Bus, which is tied to Bus 12 (138 kV) of the IEEE 14-bus system via a step-up transformer. This configuration enables the evaluation of the dynamic impact of large-scale data center loads on the transmission network and generation units. The setup reflects the realistic interconnection of data centers, which are increasingly becoming significant contributors to grid demand.

TABLE I. TERMINAL OPERATING PARAMETERS OF THE MODIFIED SYSTEM.

Buses	Voltage magnitude	Phase angle	Active power	Reactive power
Bus 1	146.30 kV	00.00°	2.324 pu	-0.166 pu
Bus 2	144.22 kV	-04.98°	0.400 pu	0.436 pu
Bus 3	139.40 kV	-12.72°	0.000 pu	0.251 pu
Bus 6	147.65 kV	-14.21°	0.000 pu	0.128 pu
Bus 8	138.00 kV	00.00°	0.250 pu	0.000 pu

The data center load is categorized into three distinct groups to capture its heterogeneous nature. The critical load (processors, primary storage, and protection systems) is modeled as mostly inductive, while the non-critical load (secondary storage and other IT equipment) is primarily resistive. In addition, a ZIP load representing constant impedance (Z), constant current (I), and constant power (P) components is considered. For the simplified load model, the critical load is set at 60 MVA, the non-critical load at 20 MVA, and the ZIP load at 20 MVA (16+j12). Furthermore, the critical and non-critical loads are modeled as dc loads interfaced with

the 11 kV Bus through an ac/dc converter, whereas the ZIP load remains connected to the bus under all conditions.

Two case studies have been designed to evaluate the system's transient response under disturbances. In Case I, the critical (60 MVA) and non-critical (20 MVA) loads are suddenly disconnected from the 11 kV Bus at 5 s and reconnected at 5.43 ms, simulating abrupt load variations. In Case II, a three-phase short-circuit fault is applied at the 11 kV Bus at 5 ms, coinciding with the disconnection of the critical and non-critical loads. It is assumed that the critical (60 MVA) loads are backed up by the local Uninterruptible Power Supply (UPS) system. The fault is cleared after 430 ms, and all loads are reconnected at 5.43 ms. These scenarios provide insight into the stability performance of the system under both load-shedding and fault-induced disturbances, highlighting the critical role of dynamic load modeling in power system studies involving large-scale data centers.

The frequency response of the system under both case studies is illustrated in Fig. 3 and Fig. 4. As observed in Fig. 3, sudden variations in the data center load significantly affect system frequency, leading to noticeable oscillations. Large-scale data center loads, due to their abrupt and dynamic nature, impose rapid changes on the system, resulting in steep rising and falling ramps in frequency. Similar behavior is evident in both cases, where the high rate of frequency deviation indicates potential instability in the system. These findings highlight the necessity of effective frequency regulation mechanisms to mitigate such high ramp in frequency changes and ensure stable and reliable system operation in the presence of large data center loads in the power system.

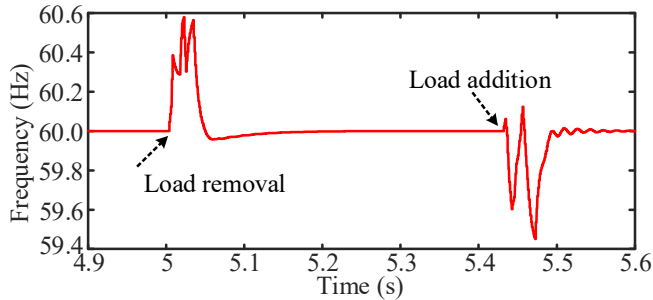


Fig. 3. System frequency response for the sudden removal and addition of load at 11 kV Bus.

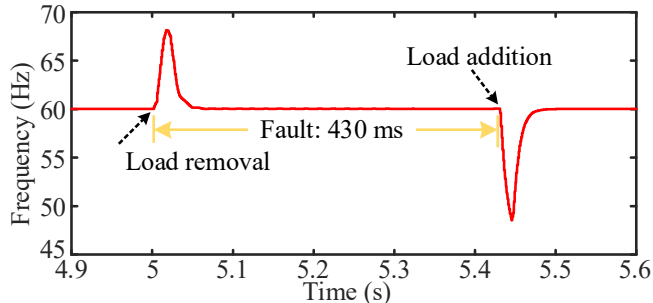


Fig. 4. System frequency response for 3-phase short-circuit faults at 11 kV Bus.

The voltage responses for Case I and Case II are illustrated in Fig. 5 and Fig. 6, respectively. Analysis of the results in Figs. 3–6 demonstrates that the sudden removal of data center load leads to an increase in both system frequency and voltage,

whereas the sudden addition of load results in a corresponding decrease. These abrupt load transitions introduce steep voltage ramps, highlighting the sensitivity of the power system to rapid changes in data center demand. Nonetheless, the observed frequency oscillations are more pronounced than the voltage oscillations, suggesting that frequency stability is more critically impacted by sudden variations in large-scale data center loads in the power system.

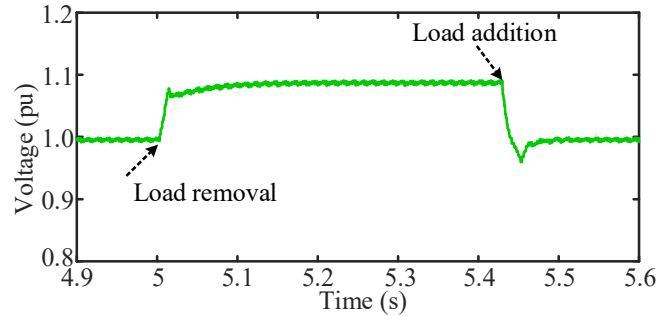


Fig. 5. Bus voltage response for the sudden removal and addition of load at 11 kV Bus.

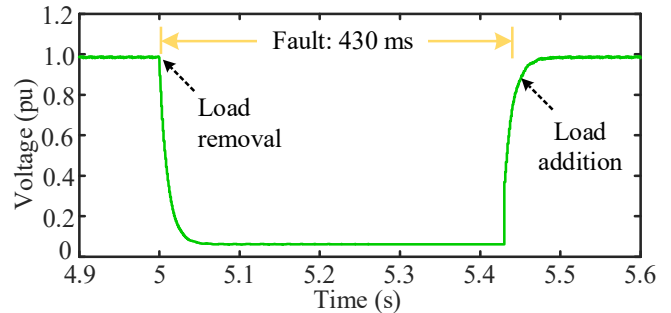


Fig. 6. Bus voltage response for 3-phase short-circuit faults at 11 kV Bus.

The data center is integrated at the 11 kV Bus, which is connected to Bus 12 of the IEEE 14-bus system. Bus 12 receives power primarily from Bus 6 and Bus 13, with the active power flows from Bus 6 to Bus 12 under both case studies illustrated in Figs. 7–10. The results indicate that sudden removal and addition of the data center load significantly affect the active and reactive power flow in the associated transmission lines. In particular, the majority of the load demand is supplied through the line between Bus 6 and Bus 12, emphasizing its critical role in supporting the data center connection and highlighting the impact of abrupt load variations on transmission line loading.

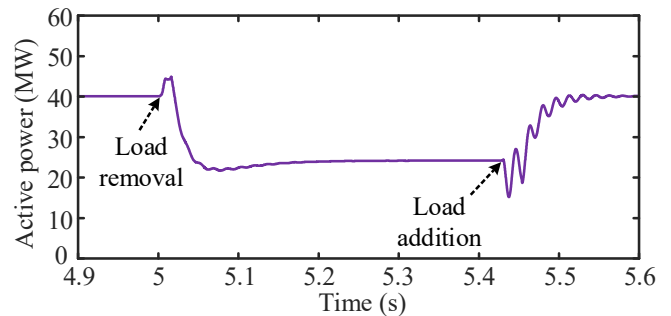


Fig. 7. Active power flow between Bus 12 and Bus 6 for the sudden removal and addition of load at 11 kV Bus.

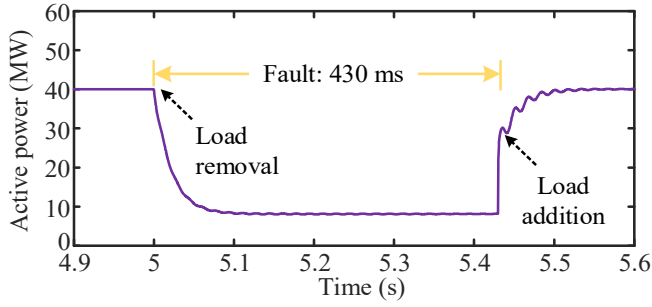


Fig. 8. Active power flow between Bus 12 and Bus 6 for 3-phase short-circuit faults at 11 kV Bus.

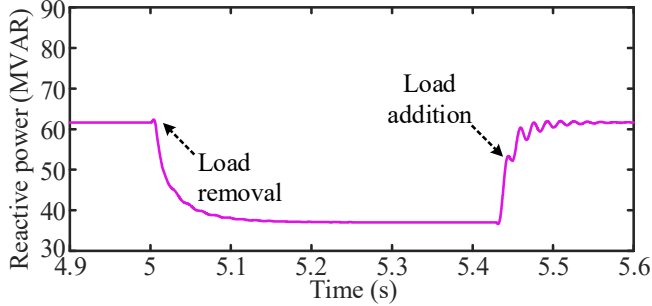


Fig. 9. Reactive power flow between Bus 12 and Bus 6 for the sudden removal and addition of load at 11 kV Bus.

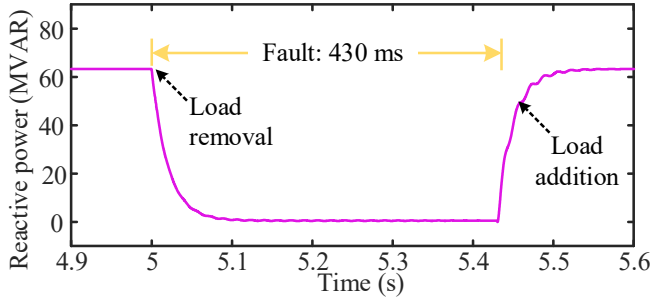


Fig. 10. Reactive power flow between Bus 12 and Bus 6 for 3-phase short-circuit faults at 11 kV Bus.

In this work, the rotor angle (δ) and speed (ω) of the synchronous generator are examined to assess the impact of data center load on its dynamic performance. The rotor angle and speed are expressed by (1) and (2), respectively.

$$\frac{d\delta}{dt} = (\omega - \omega_s) \quad (1)$$

$$\frac{d\omega}{dt} = \frac{1}{2H} (P_m - P_e - K_d(\omega - \omega_s)) \quad (2)$$

Here, ω_s , H , P_m , P_e , and K_d denote the synchronous speed, inertia constant, mechanical power, electrical power, and damping ratio of the synchronous generator, respectively.

The rotor speed oscillations resulting from the sudden addition and removal of the data center load are illustrated in Fig. 11 and Fig. 12 for Case I and Case II, respectively. As shown in Fig. 11, abrupt load variations significantly increase the amplitude of rotor speed oscillations, indicating a direct impact on generator dynamics. If not properly regulated, these oscillations may adversely affect the performance and stability of the synchronous generator, potentially leading to reduced operational reliability. Therefore, appropriate control strategies are required to mitigate the ramping behavior of rotor speed

oscillations and ensure stable generator operation under large-scale data center load disturbances.

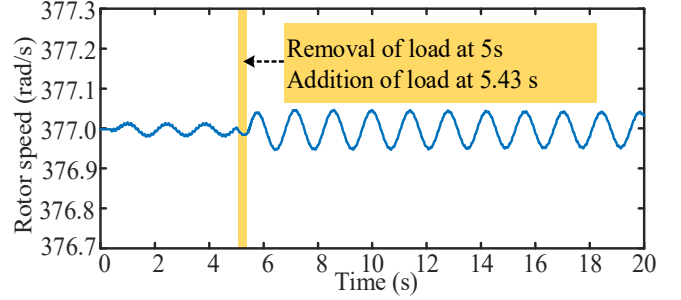


Fig. 11. Synchronous generator rotor speed for the sudden removal and addition of load at 11 kV Bus.

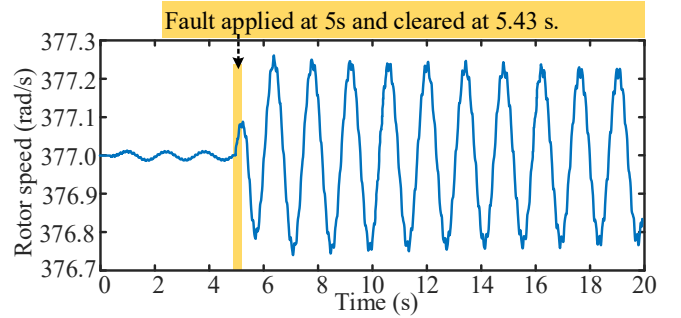


Fig. 12. Synchronous generator rotor speed for 3-phase short-circuit faults at 11 kV Bus.

IV. RECOMMENDATION FOR REGULATION STRATEGIES

To ensure the secure and reliable operation of electrical power networks under the increasing penetration of large-scale data center loads, it is essential to establish comprehensive and forward-looking regulation strategies. These regulations should address the unique operational characteristics of data centers while maintaining grid stability and performance under both normal and disturbed conditions. The following recommendations are proposed:

A. Mandate disturbance ride-through capability.

Data centers should be required to maintain operation during grid disturbances such as voltage sags, swells, and frequency deviations. Implementing disturbance ride-through capability ensures that data centers remain connected during transient events, thereby avoiding sudden large-scale load drops that could destabilize the network.

B. Require active and reactive power compensation facilities.

Regulation should enforce the provision of both active and reactive power compensation systems (e.g., static synchronous compensator (STATCOMs), synchronous condensers) at data center sites. These facilities will support voltage stability and power quality by controlling power flows and maintaining a desired power factor.

C. Protection system requirements with adequate performance margins.

Data center protection schemes should be designed with defined performance capabilities and appropriate safety margins to ensure coordination with the overall grid protection system.

This will help in isolating faults efficiently without causing unnecessary disconnections or cascading failures.

D. Pre-connection network impact assessment.

Before any data center is connected to the grid, a comprehensive grid impact study must be conducted. This study should assess whether the connection would degrade existing grid conditions, such as voltage stability, thermal limits, fault level tolerances, or sub-synchronous oscillation. Also, it is important to accurately model the data center loads for the pre-connection studies. The connection should only proceed if the study confirms no adverse impact.

E. Incorporate phase measurement unit (PMU)-based monitoring

To enhance observability and situational awareness, PMU-based monitoring systems should be integrated at points of interconnection with data centers. These devices provide high-resolution, time-synchronized measurements that help operators detect dynamic disturbances and implement corrective actions in real time.

F. Analyze active power recovery following fault clearance

The behavior of data center loads during post-fault recovery must be thoroughly examined during connection studies. Specifically, the rate and magnitude of active power recovery should be captured to ensure that sudden surges do not create secondary disturbances.

G. Define active power ramp rate limits during disturbances

Regulations should specify permissible ramp rates for active power during voltage disturbances and recovery as well for normal grid condition. This will prevent abrupt power injections or withdrawals that can stress grid components and compromise stability during dynamic conditions.

H. Validate site-specific simulation models with actual performance

A critical regulatory requirement should be the validation of data center load simulation models using on-site performance data. This model validation process ensures that planning and operational studies reflect real-world behavior, enhancing the accuracy of grid impact assessments and control strategies.

By adopting and enforcing these regulatory strategies, system operators and policymakers can better accommodate large-scale data center loads without compromising grid security, stability, or operational resilience. These recommendations support a coordinated and proactive approach to future grid planning in the context of rapidly growing digital infrastructure.

V. CONCLUSIONS

This work has demonstrated that the growing integration of large-scale data center loads into electrical power networks presents considerable challenges to grid stability and operational reliability. Through comprehensive analysis using PSCAD simulations on the IEEE 14-bus test system, it is evident that significant data center penetration can cause frequency fluctuations with high ramp, voltage instability, and increased

stress on grid infrastructure for both cases (removal and addition of loads). As data center electricity demand is expected to rise sharply, potentially reaching 321 TWh by 2030, these impacts are likely to intensify if not properly addressed. To ensure a stable and resilient power grid, it is crucial to adopt proactive strategies such as demand-side management, advanced grid control techniques, and infrastructure upgrades. Based on the findings, this work suggests valuable recommendations for utility operators, planners, and policymakers in designing robust frameworks to accommodate the accelerating load demands of future data centers while maintaining the security and efficiency of the power system. Further studies need to be carried out to understand the impact of rapid growing data center load implementations on the highly penetrated renewable energy networks operated by grid following and grid forming convert systems.

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